Chapter 2 Homework (Robotics)

```
I3 = eye(3);
syms theta phi
```

2.1 [15] A vector ${}^{A}P$ is rotated about $\hat{Z}_A$ by θ degrees and is subsequently about $\hat{X}_A$ by ϕ degrees. Give the rotation matrix that accomplishes these r in the given order.

Fixed frame rotation (left to right) -> R = Rx•Rz

```
R_x = [1 0 0; 0 cos(phi) -sin(phi); 0  sin(phi) cos(phi)]
```

R_x =

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{pmatrix}$$

```
R_z = [cos(theta)  -sin(theta) 0; sin(theta) cos(theta) 0; 0 0 1]
```

R_z =

$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

```
R = R_x * R_z
```

R =

$$\begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \cos(\phi)\sin(\theta) & \cos(\phi)\cos(\theta) & -\sin(\phi) \\ \sin(\phi)\sin(\theta) & \cos(\theta)\sin(\phi) & \cos(\phi) \end{pmatrix}$$

2.3 [16] A frame $\{B\}$ is located initially coincident with a frame $\{A\}$. We ro about $\hat{Z}_B$ by θ degrees, then we rotate the resulting frame about $\hat{X}_B$ by ϕ Give the rotation matrix that will change the descriptions of vectors fr to ${}^{A}P$.

Moving frame rotation (right to left) -> R = Rz•Rx

```
R = R_z * R_x
```

R =

$$\begin{pmatrix} \cos(\theta) & -\cos(\phi)\sin(\theta) & \sin(\phi)\sin(\theta) \\ \sin(\theta) & \cos(\phi)\cos(\theta) & -\cos(\theta)\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{pmatrix}$$

2.12 [14] A velocity vector is given by

$$^{B}V = \begin{bmatrix} 10.0 \\ 20.0 \\ 30.0 \end{bmatrix}.$$

Given

$$^{A}_{B}T = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 11.0 \\ 0.500 & 0.866 & 0.000 & -3.0 \\ 0.000 & 0.000 & 1.000 & 9.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

compute ^{A}V.

First thought: Velocity should be the same irrespective of where is "within" the frame. The fact that it's the derivative of position proves this.

```
BV = [10 20 30]'
```

```
BV = 3×1
    10
    20
    30
```

```
ARB = [sqrt(3)/2 -0.5 0; 0.5 sqrt(3)/2 0 ; 0 0 1]
```

```
ARB = 3×3
    0.8660   -0.5000         0
    0.5000    0.8660         0
         0         0    1.0000
```

```
AV = ARB * BV
```

```
AV = 3×1
   -1.3397
   22.3205
   30.0000
```

2.14 [31] Develop a general formula to obtain $^{A}_{B}T$, where, starting from init cidence, $\{B\}$ is rotated by θ about $\hat{K}$ where $\hat{K}$ passes through the point through the origin of $\{A\}$ in general).

Intuition: The rotation matrix is the Equivalent Angle-Axis Rk(θ) (from Euler's theorem). the bottom row is [0 0 0 1]. To find is the 3x1 vector describing the positional shift. This by its nature has to depend on the orientation of the axis K_hat, as well as the location of the point A_P.

Task: Find ${}^{A}P_{B,org}$ given some vector ${}^{A}P$ about which rotation ta

Given ${}^{A}P$ can we find the vector ${}^{A}P_{B,org}$?

$${}^{A}P = {}^{A}R_{B}\,{}^{B}P + {}^{A}P_{B,org}$$

$$-{}^{A}P_{B,org} = {}^{A}R_{B}\,{}^{B}P - {}^{A}P$$
$${}^{A}P_{B,org} = {}^{A}P - {}^{A}R_{B}\,{}^{B}P$$

Since ${}^{A}P$ is the center of rotation, it's necessarily equivalent to ${}^{A}P_{B,org} = {}^{A}P - {}^{A}R_{B}\,{}^{A}P$ which we can then factor out into our final

$${}^{A}P_{B,org} = (I_3 - {}^{A}R_{B})\,{}^{A}P$$

And because the rotation matrix is written as $R_k(\theta)$, we can rew

$${}^{A}P_{B,org} = (I_3 - R_k(\theta)) * {}^{A}P$$

Our final result:

```
syms R_k(theta) 3 matrix
syms Q [3 1]

R_k_theta = R_k(theta);
AP = symmatrix('AP',[3 1]);
I3 = eye(3);
ATB = [R_k_theta, (I3 - R_k_theta)*AP ; symmatrix('zeros_(1x3)',[1 3]) ,
symmatrix('ones_(1x1)',[1 1])]
```

ATB =

$$\begin{pmatrix} \boldsymbol{R_k(\theta)} & (\boldsymbol{I_3} - \boldsymbol{R_k(\theta)})\,\mathbf{AP} \\ \mathbf{zeros_{(1x3)}} & \mathrm{ones}_{(1x1)} \end{pmatrix}$$

2.20 [20] Imagine rotating a vector Q about a vector $\hat{K}$ by an amount θ to form a new vector, Q'—that is,

$$Q' = R_K(\theta)Q.$$

Use (2.80) to derive **Rodriques's formula**,

$$Q' = Q\cos\theta + \sin\theta(\hat{K} \times Q) + (1 - \cos\theta)(\hat{K} \cdot Q)\hat{K}.$$

```
syms k_x k_y k_z
syms Q [3 1]
versine = @(x) 1-cos(x);
k = [k_x ; k_y; k_z]
```

k =

$$\begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix}$$

```
k_matrix = [k_x 0 0; 0 k_y 0 ; 0 0 k_z] ;
assume(k,'real')
assume(Q,'real')
versine_component = k_matrix * [k_x k_y k_z; k_x k_y k_z ; k_x k_y k_z]
```

versine_component =

$$\begin{pmatrix} {k_x}^2 & k_x k_y & k_x k_z \\ k_x k_y & {k_y}^2 & k_y k_z \\ k_x k_z & k_y k_z & {k_z}^2 \end{pmatrix}$$

```
cosine_component = eye(3)
```

```
cosine_component = 3×3
     1     0     0
     0     1     0
     0     0     1
```

```
sine_component = [0 -k_z k_y ; k_z 0 -k_x; -k_y k_x 0]
```

sine_component =

$$\begin{pmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{pmatrix}$$

```
R_k_theta = versine(theta) * versine_component + cos(theta) *
cosine_component + sin(theta) * sine_component;
```

When taking Rk(θ)•Q, we get the following from a component breakdown (noting that matrix multiplication, as well as scalar multiplication with a matrix, are distributive):

• The cosine component is the identity matrix -> IQ = Q

• The sine component is simply the matrix form of the cross product with the vector K -> sine_component * Q = K x Q

• Q times the versine component is

```
versine_component
```

```
versine_component =
```

$$\begin{pmatrix} k_x^2 & k_x\,k_y & k_x\,k_z \\ k_x\,k_y & k_y^2 & k_y\,k_z \\ k_x\,k_z & k_y\,k_z & k_z^2 \end{pmatrix}$$

Corresponding to

```
k * k'
```

```
ans =
```

$$\begin{pmatrix} k_x^2 & k_x\,k_y & k_x\,k_z \\ k_x\,k_y & k_y^2 & k_y\,k_z \\ k_x\,k_z & k_y\,k_z & k_z^2 \end{pmatrix}$$

Which when multiplied with Q gives us

```
(k * k') * Q
```

```
ans =
```

$$\begin{pmatrix} Q_1\,k_x^2 + Q_2\,k_x\,k_y + Q_3\,k_x\,k_z \\ Q_2\,k_y^2 + Q_1\,k_x\,k_y + Q_3\,k_y\,k_z \\ Q_3\,k_z^2 + Q_1\,k_x\,k_z + Q_2\,k_y\,k_z \end{pmatrix}$$

Because matrix multiplication is associative, we can pull the brackets to the right 2 terms instead

```
k * (k' * Q)
```

```
ans =
```

$$\begin{pmatrix} k_x\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \\ k_y\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \\ k_z\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \end{pmatrix}$$

And now, because the value in the brackets is a scalar, we can put it in front of k to get closer to the original form we want (since scalar multiplication is commutative)

```
(k' * Q) * k
```

```
ans =
```

$$\begin{pmatrix} k_x\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \\ k_y\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \\ k_z\,(Q_1\,k_x + Q_2\,k_y + Q_3\,k_z) \end{pmatrix}$$

By definition, the dot product of k and Q is k_(1x3) * Q_(3x1) . Therefore, we have now proven that the versine component is equivalent to (k • Q) * k.

We can then use the distributive property of matrix multiplication to say that

Rk(θ) * Q = (v(θ)*versine_component_rkθ + cos(θ)*cosine_component_rkθ + sin(θ)*sine_component) * Q = [Rodrigues' formula] QED.

2.21 [15] For rotations sufficiently small that the approximations $\sin\theta = \theta$, $\cos\theta = 1$, and $\theta^2 = 0$ hold, derive the rotation-matrix equivalent to a rotation of θ about a general axis, $\hat{K}$. Start with (2.80) for your derivation.

cos(θ) = 1 -> v(θ) = 0

sin(θ) = θ

Remove all versine terms, and replace sin of theta with theta

```
r_k_small_theta = subs(R_k_theta,[sin(theta) cos(theta)],[theta 1])
```

```
r_k_small_theta =
```

$$\begin{pmatrix} 1 & -k_z\,\theta & k_y\,\theta \\ k_z\,\theta & 1 & -k_x\,\theta \\ -k_y\,\theta & k_x\,\theta & 1 \end{pmatrix}$$

Which can also be written as

I_3 + [K]_x * θ

2.22 [20] Using the result from Exercise 2.21, show that two infinitesimal rotations commute (i.e., the order in which the rotations are performed is not important).

Rk1 = I_3 + [K1]_x * θ

Rk2 = I_3 + [K2]_x * θ

Rk1 * Rk2 = (I_3 + [K1]_x * θ)(I_3 + [K2]_x * θ) = I_3 + [K1]_x * θ + [K2]_x * θ + [K1]_x [K2]_x * θ^2

Cancel the θ^2 term -> all other terms are scalar multiples making both orders equivalent

```
syms k1 k2 [3 1]
rk1 = subs(r_k_small_theta, k, k1)
```

```
rk1 =
```

$$\begin{pmatrix} 1 & -k_{13}\,\theta & k_{12}\,\theta \\ k_{13}\,\theta & 1 & -k_{11}\,\theta \\ -k_{12}\,\theta & k_{11}\,\theta & 1 \end{pmatrix}$$

```
rk2 = subs(r_k_small_theta, k, k2)
```

```
rk2 =
```

$$\begin{pmatrix} 1 & -k_{23}\,\theta & k_{22}\,\theta \\ k_{23}\,\theta & 1 & -k_{21}\,\theta \\ -k_{22}\,\theta & k_{21}\,\theta & 1 \end{pmatrix}$$

Subtract them from each other, replacing θ^2 with 0 -> they're the same

```
subs(rk1*rk2 - rk2 * rk1,theta^2,0)
```

```
ans =
```

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

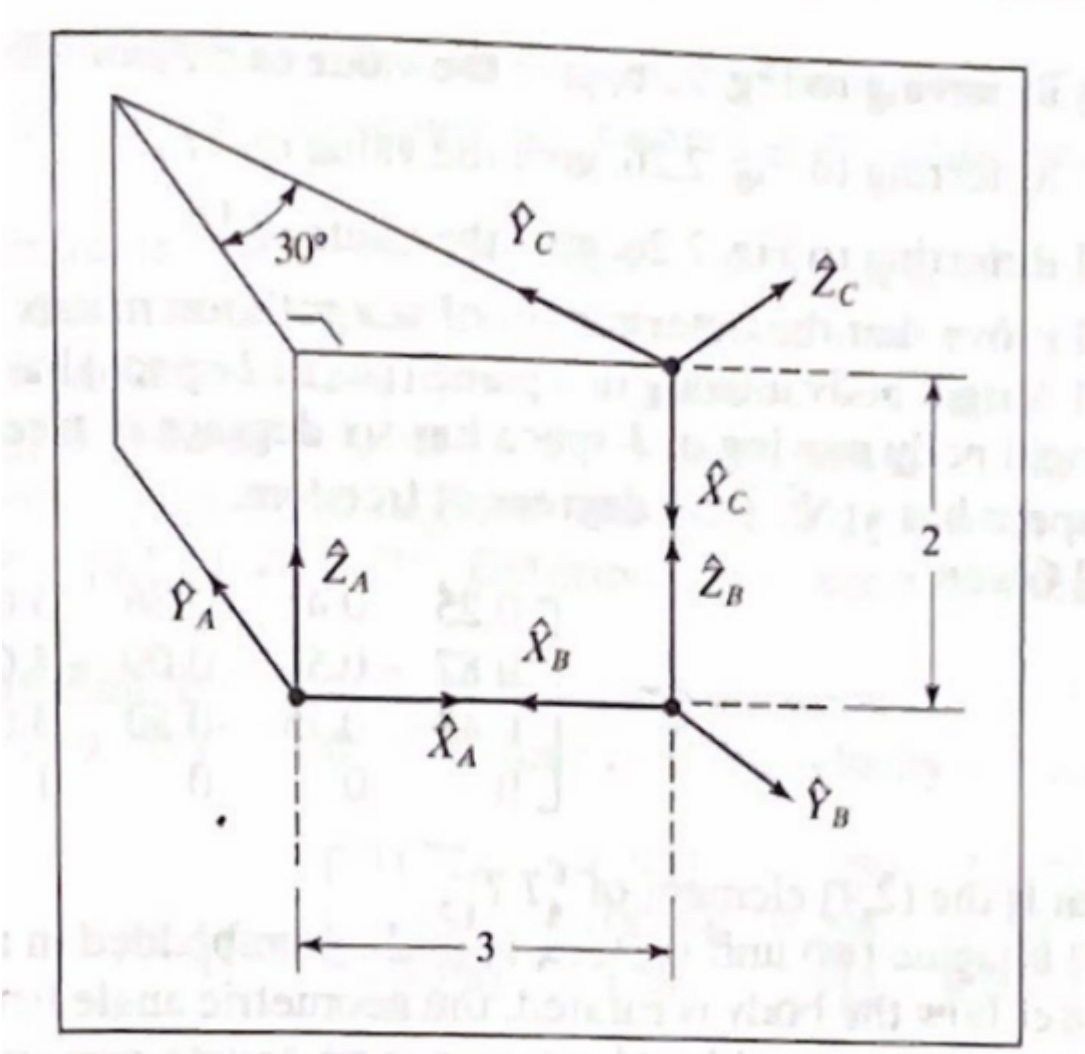

FIGURE 2.25: Frames at the corners of a wedge.

2.27 [15] Referring to Fig. 2.25, give the value of ${}^{A}_{B}T$.

Shift: +3 in the X direction of B (AP_B,org = [3 0 0]')

Y_A = -Y_B

X_A = -X_B

Z_A = Z_B

Fill in column vectors as such.

(Can also write this as a π rotation about Z, but this is simpler since we can do it visually)

```
AR_B = [-1 0 0 ; 0 -1 0 ; 0 0 1]'
```

```
AR_B = 3×3
    -1     0     0
```

```
     0    -1     0
     0     0     1
```

```
AT_B = [AR_B , [3 0 0]' ; 0 0 0 1]
```

```
AT_B = 4×4
    -1     0     0     3
     0    -1     0     0
     0     0     1     0
     0     0     0     1
```

2.37 [15] Given

$$ {}^{A}_{B}T = \begin{bmatrix} 0.25 & 0.43 & 0.86 & 5.0 \\ 0.87 & -0.50 & 0.00 & -4.0 \\ 0.43 & 0.75 & -0.50 & 3.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, $$

what is the (2,4) element of ${}^{B}_{A}T$?

```
AT_B = [0.25 0.43 0.86 5.0; 0.87 -0.5 0 -4; 0.43 0.75 -0.5 3 ; 0 0 0 1]
```

```
AT_B = 4×4
    0.2500    0.4300    0.8600    5.0000
    0.8700   -0.5000         0   -4.0000
    0.4300    0.7500   -0.5000    3.0000
         0         0         0    1.0000
```

```
AR_B = AT_B(1:3,1:3)
```

```
AR_B = 3×3
    0.2500    0.4300    0.8600
    0.8700   -0.5000         0
    0.4300    0.7500   -0.5000
```

```
BT_A = [AR_B , -AR_B' * AT_B(1:3,4); 0 0 0 1]
```

```
BT_A = 4×4
    0.2500    0.4300    0.8600    0.9400
    0.8700   -0.5000         0   -6.4000
    0.4300    0.7500   -0.5000   -2.8000
         0         0         0    1.0000
```

```
BT_A(2,4)
```

```
ans =
-6.4000
```

The answer is -6.4

2.38 [25] Imagine two unit vectors, v_1 and v_2, embedded in a rigid body. Note that, no matter how the body is rotated, the geometric angle between these two vectors is preserved (i.e., rigid-body rotation is an "angle-preserving" operation). Use this fact to give a concise (four- or five-line) proof that the inverse of a rotation matrix must equal its transpose and that a rotation matrix is orthonormal.

For some rotation R, we are given that the angle is preserved between vectors. This means that the dot product is also preserved since they are both unit vectors.

This means, $v1 \cdot v2 = Rv1 \cdot Rv2$, equivalent to $(Rv1)^T * Rv2$

since $(AB)^T = B^T A^T$ we can rewrite this as $v1^T R^T R v2$

For this to hold true, $R^T * R$ must be equal to I. In other words, R's transpose must also be its inverse. QED.